

Alveolar Epithelial Cells in Lethal COVID-19 Fan Out Along a Continuous Injury Axis

Trajectory Analysis of SCP1219 with Ten-Way Ablation Robustness

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Class Project

What we asked

- Do alveolar epithelial cells in lethal COVID-19 fail as a coherent block, or as a continuous graded process?
- Pre-registered concept: *epithelial robustness collapse*.
- Three pre-specified criteria:
 1. Centroid displacement (one-sided Mann–Whitney U , COVID $>$ Healthy)
 2. Within-group dispersion (Levene)
 3. Pseudotime enrichment (1000-permutation test on median difference)

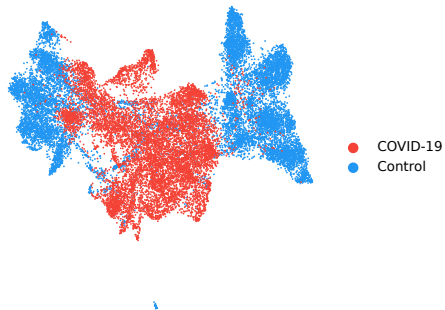
- Columbia/NYP COVID-19 Lung Atlas (SCP1219; Melms et al. 2021)
- 116,313 snRNA-seq nuclei → retained AT1/AT2/ECM-high epithelial → **22,128 cells**
- 9,608 AT1; 11,341 AT2; 1,179 ECM-high transitional
- 7 healthy (12,181 nuclei) + 20 COVID-19 donors (9,947 nuclei)
- Per-donor counts: 62 to 2,850 (10/27 donors < 300 cells)

- Scanpy 1.12.1 / anndata 0.12.10 / Python 3.12.3, seed 42
- Normalize $\rightarrow$ log1p $\rightarrow$ 3,000 HVGs (Seurat v3) $\rightarrow$ scale $\rightarrow$ 50-PC PCA
- k -NN ($k=15$) $\rightarrow$ UMAP, diffusion map (15 comp.)
- Diffusion pseudotime rooted at healthy-AT2 centroid, normalized to $[0, 1]$
- 8 curated gene programs scored with `sc.tl.score_genes`
- 10 pre-specified ablations

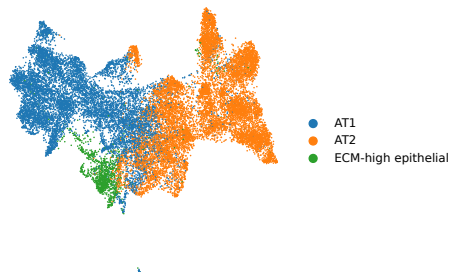
Harmony failed at runtime (harmonypy returned shape (50,) instead of (22128,50)); pipeline fell back to uncorrected PCA.

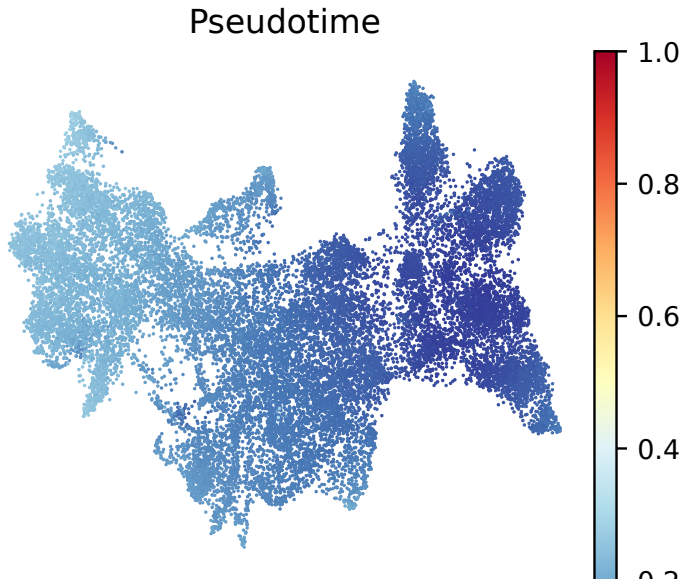
The manifold

Condition



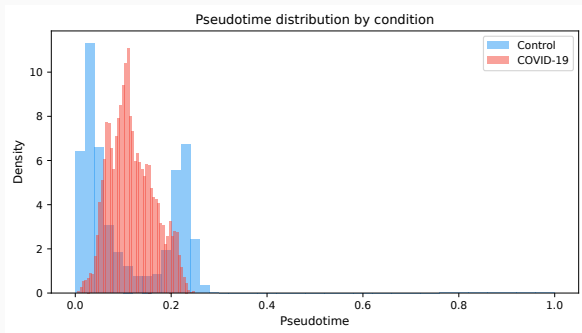
Cell type





Result 1 — pseudotime shift

- Control median 0.063; COVID median 0.111; observed difference +0.0483
- 1,000-permutation null: mean 6.0×10^{-5} ; std 1.24×10^{-3}
- Observed never exceeded in null $\Rightarrow p = 0.000999$
- LOO (27 iters): direction preserved in **all 27**; range 0.028–0.202
- Magnitude is modest (4.8% of the normalized axis).



Result 2 — centroid displacement NOT supported

- One-sided Mann–Whitney U in PCA-50 (COVID > Healthy): $U = 5.21 \times 10^7$, $p = 1.0$
- Means: COVID 12.36 vs. Healthy 12.10
- Medians: COVID **10.68** vs. Healthy **11.56** — COVID median is *lower*
- Rank-biserial $r = 0.14$ points opposite to the alternative

Pre-specified core test **failed**. We report this prominently.

Result 3 — dispersion IS supported

- Within-group distance to own centroid, PCA-50
- Variance: COVID **33.28** vs. Healthy **14.58** (2.3-fold)
- Levene: statistic 276.5; $p < 10^{-30}$

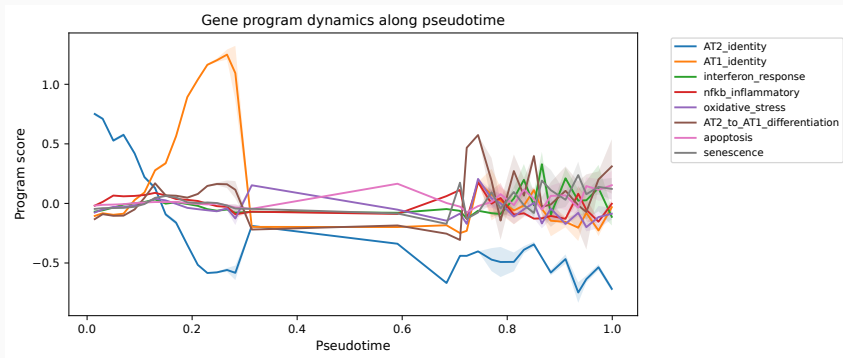
COVID-19 cells are *spread out*, not *pushed away*.

Program ordering along pseudotime

Rank	Program	Peak	Peak score
1	AT2 identity	0.015	0.75
2	AT1 identity	0.266	1.25
3	Apoptosis	0.591	0.17
4	AT2→AT1 diff.	0.744	0.57
4 (tie)	NF- κ B	0.744	0.18
4 (tie)	Oxidative stress	0.744	0.21
7	Interferon	0.866	0.33
8	Senescence	0.935	0.24

Not the hypothesized IFN→NF- κ B→repair→apoptosis cascade. Apoptosis peaks *before* inflammatory/stress.

Program dynamics



KRT8⁺ transitional cells

- 1,179 ECM-high / KRT8⁺/CLDN4⁺ transitional cells
- **3.1× enriched in COVID-19:** 8.5% vs. 2.7%; χ^2 $p < 10^{-30}$
- Intermediate pseudotime (mean 0.3–0.5)
- Concordant with DATP / aberrant-basaloid literature (Kobayashi 2020; Strunz 2020; Adams 2020; Habermann 2020)
- Correlational: snapshot, not lineage trace

Ten ablations — summary

Ablation	MPD	ρ	r	Notes
1 AT2-only	+0.133	0.33	−0.29	sign flip on r
2 UMAP / diffmap	+0.048	0.55	+0.14	unchanged
3 Harmony*	+0.048	0.55	+0.14	both branches identical (Harmony failed)
4 No apoptosis genes	+0.074	0.50	+0.14	slight up
5 4 roots	−0.037 to +0.048	−0.10 to 0.64	+0.14	root-dependent
6 Broader epithelial	−0.008	0.83	+0.25	shift abolished
7 LOO (27)	0.028–0.202	−0.17–0.74	per-iter	direction preserved 27/27
8 Curated-only	+0.048	1.00	+0.14	stable
8 MSigDB-only	+0.048	−0.80	+0.14	ordering inverted
9 Palantir	—	—	—	not installed
10 Permutation	$p = 0.001$	—	—	1000 shuffles

Composite assessment

Criterion	Outcome
(1) Centroid displacement ($p < 0.05$)	NOT supported ($p = 1.0$)
(2) Dispersion (Levene $p < 0.05$)	supported ($p \approx 0$)
(3) Pseudotime enrichment (perm. $p < 0.05$)	supported ($p = 0.001$)
Composite: <i>moderate</i> (2 of 3)	

What the data actually support

- Lethal COVID-19 *fans out* alveolar epithelium along a continuous pseudotime axis.
- Signal is **loss of coherence**, not directional block displacement.
- KRT8⁺/CLDN4⁺ transitional population accumulates at intermediate pseudotime (3.1× enriched) — consistent with stalled AT2→AT1 repair.
- The pseudotime shift is robust. The fine program ordering is *hypothesis-generating*.

What the data do NOT support

- Centroid displacement (pre-specified core test; $p = 1.0$).
- A specific IFN $\rightarrow$ NF- κ B $\rightarrow$ repair $\rightarrow$ apoptosis cascade (ordering inverts under MSigDB or COVID-extreme root).
- A pan-epithelial lung trajectory — adding airway cells (Ablation 6) abolishes the shift.
- Any claim about elapsed time — pseudotime is a manifold ordering, not a clock.

Known limitations

- Cross-sectional, post-mortem, single dataset, lethal cases only.
- Donor imbalance: 10/27 donors < 300 cells.
- Harmony failed $\Rightarrow$ Ablation 3 cannot distinguish corrected vs. uncorrected.
- Palantir not installed $\Rightarrow$ Ablation 9 no alt-trajectory comparison.
- GPX1 absent from HVGs (oxidative stress dropout); other programs unchecked.
- Fine ordering of injury programs sensitive to gene-list choice.

Takeaways

1. Modest but robust pseudotime shift ($+0.048$, $p = 0.001$).
2. Significant dispersion increase ($2.3\times$ variance).
3. Centroid displacement does not distinguish the conditions.
4. KRT8⁺ transitional population $3.1\times$ enriched at intermediate pseudotime.
5. Future robustness-collapse studies: use dispersion and trajectory metrics, not centroid displacement.

Questions?